

Operator Norm Definition

$$\|A\|_B = \inf \{M : \|A(x)\| \leq M\|x\|, \forall x \in X\}$$

Norm of bounded linear operator $A: X \rightarrow Y$

Equivalent forms:

- $\|A\|_B = \sup \{\|A(x)\| : \|x\| = 1\}$
- $\|A\|_B = \sup \{\|A(x)\|/\|x\| : x \neq 0\}$
- $\|A\|_B = \sup \{\|A(x)\| : \|x\| \leq 1\}$

Properties of operator norm:

1. $\|A(x)\| \leq \|A\|_B \cdot \|x\|$ for all $x \in X$
2. $\|A + B\|_B \leq \|A\|_B + \|B\|_B$ (triangle inequality)
3. $\|\lambda A\|_B = |\lambda| \cdot \|A\|_B$ for scalar λ
4. $\|A \circ B\|_B \leq \|A\|_B \cdot \|B\|_B$ (composition)